

Beyond “Arab” as a Single Category: A Counterfactual Audit of National-Origin Bias in LLM Hiring

Framing outline: research question, abstract, introduction

Internal working notes, 2026-10-04

AI-assisted (Claude). Notes, not paper text.

The team writes the abstract and the introduction in its own words.

Abstract

Not an abstract. The seven slots an abstract has to fill, as notes. ^F = full text read, ^A = abstract only.

- **1 Context.** LLMs proposed for CV screening. Audits so far: mostly US race and gender, signalled by names.
- **2 Motivation: the documented effects.**
 - Human employers, Western labour markets: about 41% fewer callbacks for Arab, North African and Middle Eastern names (pooled ratio 0.59) (Lippens et al., 2023b)^F.
 - LLMs, generated text: Muslims tied to violence in 66% of GPT-3 completions (Abid et al., 2021)^A. Arabs, not Westerners, named the “loser group” in 79% of model-by-topic cells (Saeed et al., 2024)^A. MENA names 78% under-represented in generated workplace stories (Shieh et al., 2026)^F.
 - LLMs, hiring: Arab names –1.41 points on a 1–100 scale, selection ratio 0.85, GPT-3.5 (Lippens, 2024)^F. GPT-4 sends Arabic names to the lower-status job more often (Bai et al., 2025)^A.
 - Newer models: one null, some reversals. So the hiring effect is open, which is a reason to test it.
- **3 Why the Arab world.** Nationality is an open sorting criterion in these labour markets. Qatar, low-income workers: Egyptian QR1,454 vs Nepali QR853 a month (Gardner et al., 2013)^F. Jordan: segmentation “based not on education or initial skills but rather on nationality” (Razzaz, 2017)^F. Gulf states expelled workers by Arab nationality in 1990–91 (Kapiszewski, 2006)^F. LLM recruiting tools are being built for this market (Albaroudi et al., 2026)^F. Limit: differences among Arab nationalities at equal qualifications are barely studied.
- **4 Gap.** “Arab” is always one pooled group. No LLM hiring study found that varies indi-

vidual Arab nationalities with qualifications fixed (tentative).

- **5 What we do.** Identical synthetic CVs; only the `Nationality`: line changes. 22 Arab League nationalities, 3 benchmarks (German, Polish, Turkish), not stated, 8 placebo nationalities. CVs and jobs set in 8 Arab countries. Open-weight models. Baseline vs neutrality instruction; independent scoring and forced choice. Outcomes: interview yes/no, fit score 0–100.
- **6 Results.** Empty until real data exist. One label per model: meaningful spread, trivial spread, undetermined, null, inconclusive.
- **7 Takeaway.** Decided by the data. Either: spread among Arab nationalities exceeds noise and placebo spread, so the pooled category hides differences. Or: bounded null, a single category is adequate for these models.

1 Research question

- **Working wording** (project brief; the team finalises it): “Do LLMs used as simulated recruiters evaluate otherwise-identical synthetic applicants differently when only the applicant’s stated national origin changes, and does a single ‘Arab’/‘MENA’ category hide systematic differences among individual Arab national origins?”
- **Primary, RQ1.** Same CV, 22 Arab League nationality signals. Is there spread? How large against a smallest effect of interest (1 point on the 0–100 fit score)? Larger than the spread among 8 unrelated placebo nationalities? The same across models?
- **Secondary.** RQ2: Arab average vs German, Polish, Turkish, not stated. RQ3: effect of a neutrality instruction. RQ4 (exploratory): independent scoring vs forced choice. SQ1: stated principle vs behaviour.
- **The argument in one line.** Known: a penalty for the pooled group. Pooled because, from outside, the origins look alike (shared names,

Premise	What the check found	Main sources
Human employers penalise Arab / MENA names	Holds for Western labour markets. Large on average, very uneven, some nulls.	Lippens et al., 2023b; Quillian and Lee, 2023; Arnoult et al., 2021
The same inside Arab labour markets	No field experiment found. Audits there vary gender only.	Krafft, 2025
LLMs penalise Arab applicants in hiring decisions	Weak and mixed: one small penalty, one null, reversals in newer models.	Lippens, 2024; Hoffmann et al., 2026; Bunel et al., 2026
LLMs stereotype Arabs and Muslims in generated text	Holds, but mostly 2020–2023 models, and pooled.	Abid et al., 2021; Shieh et al., 2026
Prior work pools Arab / MENA	Holds in both literatures.	Lippens et al., 2023b; Lippens, 2024
Names cannot tell Arab origins apart	Holds for human raters, on thin evidence (one study, one Arab origin).	Martiniello and Verhaeghe, 2022
Humans treat Arab origins differently	Mixed. A few pairs in Western settings; inside the Arab world mostly one group.	Vernby and Dancygier, 2019; Shockley and Gengler, 2024; Blaydes and Gengler, 2023
An LLM hiring study splits Arab nationalities	None found.	Albaroudi et al., 2026 (pooled)

Table 1: Literature check of 2026-10-04: what each premise of the introduction can carry. 68 sources added; details in `research/literature_review.md`, section 6.

language, religion). Open: does a model still separate them when the only cue is the nationality itself?

- **Terms.** Our set is the 22 Arab League nationalities, not “MENA”. MENA in cited work is a different set: it includes Turkish applicants in Quillian and Lee (2023)^F, and Iranian and Israeli respondents in the US census test, where Somali and Sudanese respondents mostly do not take the label (Mathews et al., 2017)^F. Report prior work under its own label.

2 Introduction: paragraph plan

Each paragraph: the point, the evidence, and what not to overstate. Table 1 sums up what each premise can carry.

P1. LLMs in hiring; what audits cover.

- Point: many LLM screening audits; almost all vary race and gender by name.
- Evidence: about 361,000 résumés, five models (An et al., 2025)^A; Gaebler et al. (2024)^F; Armstrong et al. (2024)^F. Nationality in 13.2% of 189 LLM-bias papers, gender in 79.9% (Ghosh and Wilson, 2025)^A (not hiring-specific).
- Careful: direction is unstable; newer models often show no gap or the reverse (Gao et al., 2026)^A.

P2. The premise: human employers.

- Point: discrimination against applicants with Arab, North African or Middle Eastern names is well documented and large on average, in Western labour markets.

- Evidence: meta-analysis, pooled Arab/Maghrebi/Middle Eastern callback ratio 0.59 [0.55, 0.64], 31 effects (Lippens et al., 2023b)^F. Higher after 2000 than in the 1990s, six Western countries (Quillian and Lee, 2023)^F. France: 22.8% vs 33.3% callbacks, 9,600 applications (Arnoult et al., 2021)^F (grey literature). Netherlands: 31% vs 46% (Thijssen et al., 2021)^F. Australia: 22% vs 35% (Booth et al., 2012)^F. Sweden: Iraq-born 10% vs 21% (Vernby and Dancygier, 2019)^F.
- Careful: (a) all Western; no field experiment with applicant nationality found inside an Arab country. (b) Sizes vary a lot ($I^2 = 87\%$) and there are nulls (Challe et al., 2024; Baert et al., 2017). (c) Not always the lowest group: Chinese names in Booth et al. (2012), Eastern European in Lippens et al. (2023a). (d) The signal is almost always a name. So: “well documented in Western labour markets”, not “everywhere” and not “the most”.

P3. The premise: LLMs.

- Point: stereotypes in generated text are documented; evidence on decisions is sparse and mixed.
- Evidence, text: Muslim–violence completions in GPT-3 (Abid et al., 2021)^A, persisting with names in later GPT models (Hemmatian et al., 2023)^F. MENA names 78% under-represented in generated workplace stories (Shieh et al., 2026)^F. Middle-Eastern personas conflated with religion (Cheng et al., 2023)^F.

- Evidence, decisions: GPT-3.5, pooled Arab names, −1.41 points on a 1–100 scale, ratio 0.85 at a cut-off (Lippens, 2024)^F. GPT-4 sends Arabic/Muslim names to the lower-status job more often (Bai et al., 2025)^A (detail read in the arXiv version). Gemini 2.0 Flash, pooled Arabic male names: about zero (Hoffmann et al., 2026)^F.
- Against: 18 LLMs recommend less discrimination against a real origin (Moroccan) than against a fictional group (Bunel et al., 2026)^F. Every race or ethnicity bias detected in hiring, loans and admissions favours the minority (Arcuschin et al., 2026)^F.
- Careful: do not write “LLMs discriminate against Arab applicants” as settled. Newer models reverse most where the group is named explicitly, which is our cue. Expectations stay two-sided.

P4. The single category.

- Point: both literatures pool. Pooling is natural, because from outside the origins are hard to tell apart.
- Evidence, pooling: raters assign 34% of Moroccan names to Morocco (Martiniello and Verhaeghe, 2022)^F. A standard name classifier puts eleven Arab countries in one class (Ye et al., 2017)^F. Meta-analyses pool (Lippens et al., 2023b; Quillian and Lee, 2023); so do Gulf wage studies (Alfarhan and Al-Busaidi, 2019)^A.
- Evidence that origin can matter (humans): Germany, Iraqi and Moroccan below Turkish (Koopmans et al., 2019)^F. Origin-country traits predict callbacks across 22 origins (Di Stasio and de Vries, 2024)^A. Sweden: Iraq 10% vs Somalia 5% (Vernby and Dancygier, 2019)^F. Gulf state policy: expulsions by Arab nationality, 1990–91 (Kapiszewski, 2006)^F (grey literature).
- Evidence that it may not (humans, inside the Arab world): no significant nationality differences among Qatari citizens (Shockley and Gengler, 2024)^F. Arab expatriates in Kuwait and Qatar treat “fellow Arabs” as one group (Blaydes and Gengler, 2023)^F.
- LLMs outside hiring do separate Arab countries: values, 14 Arab states (Zahraei et al., 2025)^F; sentiment by demonym (Narayanan Venkit et al., 2023)^F. The critique of “one Arab culture” (Kegel, 2025)^A.
- Careful: the human evidence points both ways, so the question is open in both directions. The

Iraq–Somalia gap mixes origin with race. Outcomes in the two Gulf studies are residency and migration attitudes, not hiring.

P5. Why the Arab world.

- Point: there, nationality openly sorts workers, so a nationality line on a CV is realistic and what a model does with it matters.
- Evidence: pay by nationality among low-income workers in Qatar, Egyptian QR1,454 vs Nepali QR853 a month (Gardner et al., 2013)^F (plain averages, no controls). Jordan: segmentation by nationality, strong demand for Egyptian workers (Razzaz, 2017)^F (ILO report). Gulf: Arab share of foreign workers from 72% (1975) to 32% (2002/4); expulsions by Arab nationality in 1990–91 (Kapiszewski, 2006)^F (UN paper). Kuwait, nine nationality groups: about a third of earnings inequality left unexplained (Al-Qudsi and Shah, 1991)^A. An LLM recruiting pipeline for the Saudi market (Albaroudi et al., 2026)^F.
- Careful: most of this compares Arab vs Asian vs Western workers, or is state policy. Differences among Arab nationalities at equal qualifications are barely studied; no field experiment exists. Say “nationality matters in these labour markets”, not “Arab nationalities are known to be treated differently”. Three of the sources are reports, not journal articles.

P6. Gap and this paper.

- Gap: no study found in which an LLM makes a hiring decision and national origin varies across more than one Arab country (searches of 29–30.09 and 04.10.2026; coverage in the literature review). Nearest: 18 Arab nationalities pooled into Arab vs non-Arab (Albaroudi et al., 2026)^F. Wording: “to our knowledge”.
- Design, in one breath: see abstract slot 5.
- What “systematic difference” means here: above decoder noise (permutation test), above the spread of placebo nationalities, and still there without Somalia, Djibouti and Comoros.
- Why a nationality line and not names: names cannot carry the country (P4).

P7. Contribution and scope.

- Contribution candidates (team wording; novelty is tentative): first split of the Arab category by nationality in LLM hiring decisions; spread judged against noise and placebo spread; effect of a neutrality instruction on the spread, not only on the mean.

- Not claimed as new: that LLMs penalise Arab applicants; that LLMs tell Arab countries apart in general; that origin matters in human hiring.
- Scope: an audit, not a hiring tool. Synthetic applicants and employers. Explicit nationality cue only, no names. The fit score is not a hiring probability. Open-weight models at pinned revisions.

3 Open points for the team

- **Western premise, Arab-world setting.** All hiring field experiments on Arab applicants are from Europe, North America, Australia or Israel. Our CVs and jobs are in eight Arab countries. The bridge is P5 (nationality openly sorts workers there). A second one, if wanted: Western defaults of LLMs in Arab contexts (Naous et al., 2024)^A. Do not present the Western callback gap as evidence about Arab labour markets.
- **A null is a live outcome.** Inside the Arab world the human evidence leans to one group. “A single category is adequate for these models” would agree with it. Write P4 so that either result reads as an answer.
- **Novelty.** Tentative. One paper is still unread in full (Bilon, 2025)^A: its abstract reports only U.S. vs non-U.S.
- **One withdrawn source.** MIRAGE (Mohammad and Sheikh 2026): its numbers are placeholders. Do not cite them.
- **Before any number goes into the paper:** every source marked ^A needs a full-text check.

References

- Abubakar Abid, Maheen Farooqi, and James Zou. 2021. [Persistent anti-muslim bias in large language models](#). In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pages 298–306.
- Sulayman S. Al-Qudsi and Nasra M. Shah. 1991. [The relative economic progress of male foreign workers in Kuwait](#). *International Migration Review*, 25(1):141–166.
- Elham Albaroudi, Taha Mansouri, Mohammad Hatamleh, and Ali Alameer. 2026. [Addressing intersectional bias in AI recruitment using HITHIRE model: a fair, ethical, green AI and transparent hiring solution for Saudi Arabia’s diverse workforce in line with vision 2030](#). *AI and Ethics*, 6(1):57.
- Usamah F. Alfarhan and Samir Al-Busaidi. 2019. [Migrant earnings gaps in Gulf Cooperation Council countries: Employers’ perceptions or opportunity costs?](#) *International Labour Review*, 158(2):273–295.
- Jiafu An, Difang Huang, Chen Lin, and Mingzhu Tai. 2025. [Measuring gender and racial biases in large language models: Intersectional evidence from automated resume evaluation](#). *PNAS Nexus*, 4(3):pgaf089.
- Iván Arcuschin, David Chanin, Adrià Garriga-Alonso, and Oana-Maria Camburu. 2026. [Biases in the blind spot: Detecting what LLMs fail to mention](#). In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*.
- Lena Armstrong, Abbey Liu, Stephen MacNeil, and Danaë Metaxa. 2024. [The silicon ceiling: Auditing GPT’s race and gender biases in hiring](#). In *Proceedings of the 4th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO ’24)*, pages 1–18.
- Émilie Arnoult, Marie Ruault, Emmanuel Valat, Pierre Villedieu, Thomas Breda, Nicolas Jacquemet, Morgane Laouénan, Roland Rathelot, Mirna Safi, Joyce Sultan Parraud, Amélie Allegre, Anna Bagramova, Frédérique Bouvier, Fabrice Foroni, Sara Ftoh Fennane, Isabelle Huet, Bianka Kozma, Amine Medaghri Alaoui, and Elshaday Tekle. 2021. [Discrimination à l’embauche des personnes d’origine supposée maghrébine : quels enseignements d’une grande étude par testing ?](#) Notes IPP 76, Institut des politiques publiques (IPP), Dares and ISM Corum.
- Stijn Baert, Andrea Albanese, Sofie du Gardein, Jolien Ovaere, and Jarno Stappers. 2017. [Does work experience mitigate discrimination?](#) *Economics Letters*, 155:35–38.
- Xuechunzi Bai, Angelina Wang, Ilia Sucholutsky, and Thomas L. Griffiths. 2025. [Explicitly unbiased large language models still form biased associations](#). *Proceedings of the National Academy of Sciences*, 122(8):e2416228122.
- Xavier Javines Bilon. 2025. [Sociodemographic biases in employment decision-making by large language models: experimental evidence from ChatGPT simulations](#). *Journal of Decision Systems*, 34(1):2599912.
- Lisa Blaydes and Justin Gengler. 2023. [Arab identity and attitudes toward migration in Kuwait and Qatar](#). *Research & Politics*, 10(4).
- Alison L. Booth, Andrew Leigh, and Elena Varganova. 2012. [Does ethnic discrimination vary across minority groups? evidence from a field experiment](#). *Oxford Bulletin of Economics and Statistics*, 74(4):547–573.
- Matthieu Bunel, Marie-Noëlle Lefebvre, and Elisabeth Tovar. 2026. [The customer is always right? how LLMs and humans judge discrimination](#). Working Paper 2026-16, EconomiX, Université Paris Nanterre.

- Laetitia Challe, Yannick L'Horty, Pascale Petit, and François-Charles Wolff. 2024. [Cyclical behavior of hiring discrimination: evidence from repeated experiments in France](#). *The Annals of Regional Science*, 72(3):711–733.
- Myra Cheng, Esin Durmus, and Dan Jurafsky. 2023. [Marked personas: Using natural language prompts to measure stereotypes in language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1504–1532.
- Valentina Di Stasio and Anne Margaretha de Vries. 2024. [Same religion, different treatment. the role of origin country characteristics in employers' decisions to hire Muslims](#). *Journal of Ethnic and Migration Studies*, 50(10):2444–2467.
- Johann D. Gaebler, Sharad Goel, Aziz Z. Huq, and Prasanna B. Tambe. 2024. [Auditing large language models for race & gender disparities: Implications for artificial intelligence-based hiring](#). *Behavioral Science & Policy*, 10(2):46–55.
- Zhenyu Gao, Wenxi Jiang, and Yutong Yan. 2026. [Can LLMs hire fairly? racial bias in resume screening](#). *Preprint*, arXiv:2606.28978.
- Andrew Gardner, Silvia Pessoa, Abdoulaye Diop, Kaltham Al-Ghanim, Kien Le Trung, and Laura Harkness. 2013. [A portrait of low-income migrants in contemporary Qatar](#). *Journal of Arabian Studies*, 3(1):1–17.
- Sourojit Ghosh and Kyra Wilson. 2025. [Bias is a math problem, AI bias is a technical problem: 10-year literature review of AI/LLM bias research reveals narrow \[gender-centric\] conceptions of 'bias', and academia-industry gap](#). *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 8(2):1091–1106.
- Babak Hemmatian, Razan Baltaji, and Lav R. Varshney. 2023. [Muslim-violence bias persists in debiased GPT models](#). *Preprint*, arXiv:2310.18368.
- Morgane Hoffmann, Emma Jouffroy, Warren Jouanneau, Marc Palyart, and Charles Pebereau. 2026. [Evaluating LLM behavior in hiring: Implicit weights, fairness across groups, and alignment with human preferences](#). *Preprint*, arXiv:2601.11379.
- Andrzej Kapiszewski. 2006. [Arab versus Asian migrant workers in the GCC countries](#). Technical Report UN/POP/EGM/2006/02, United Nations Population Division, Department of Economic and Social Affairs, Beirut.
- Amr Keleg. 2025. [LLM alignment for the arabs: A homogenous culture or diverse ones](#). In *Proceedings of the 3rd Workshop on Cross-Cultural Considerations in NLP (C3NLP 2025)*, pages 1–9.
- Ruud Koopmans, Susanne Veit, and Ruta Yemane. 2019. [Taste or statistics? a correspondence study of ethnic, racial and religious labour market discrimination in Germany](#). *Ethnic and Racial Studies*, 42(16):233–252.
- Caroline Krafft. 2025. [Do employers discriminate against married women? evidence from a field experiment in Egypt](#). *Journal of Development Economics*, 174:103452.
- Louis Lippens. 2024. [Computer says 'no': Exploring systemic bias in ChatGPT using an audit approach](#). *Computers in Human Behavior: Artificial Humans*, 2(1):100054.
- Louis Lippens, Axana Dalle, Fanny D'hondt, Pieter-Paul Verhaeghe, and Stijn Baert. 2023a. [Understanding ethnic hiring discrimination: A contextual analysis of experimental evidence](#). *Labour Economics*, 85:102453.
- Louis Lippens, Siel Vermeiren, and Stijn Baert. 2023b. [The state of hiring discrimination: A meta-analysis of \(almost\) all recent correspondence experiments](#). *European Economic Review*, 151:104315.
- Billie Martiniello and Pieter-Paul Verhaeghe. 2022. [Signaling ethnic-national origin through names? the perception of names from an intersectional perspective](#). *PLoS ONE*, 17(8):e0270990.
- Kelly Mathews, Jessica Phelan, Nicholas A. Jones, Sarah Konya, Rachel Marks, Beverly M. Pratt, Julia Coombs, and Michael Bentley. 2017. [2015 National Content Test: Race and ethnicity analysis report](#). Technical report, U.S. Census Bureau, Washington, DC.
- Tarek Naous, Michael J. Ryan, Alan Ritter, and Wei Xu. 2024. [Having beer after prayer? measuring cultural bias in large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16366–16393.
- Pranav Narayanan Venkit, Sanjana Gautam, Ruchi Panchanadikar, Ting-Hao Huang, and Shomir Wilson. 2023. [Nationality bias in text generation](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 116–122.
- Lincoln Quillian and John J. Lee. 2023. [Trends in racial and ethnic discrimination in hiring in six Western countries](#). *Proceedings of the National Academy of Sciences*, 120(6):e2212875120.
- Susan Razzaz. 2017. [A challenging market becomes more challenging: Jordanian workers, migrant workers and refugees in the Jordanian labour market](#). Technical report, International Labour Organization, Regional Office for Arab States, Beirut.
- Muhammed Saeed, Elgizouli Mohamed, Mukhtar Mohamed, Shaina Raza, Muhammad Abdul-Mageed, and Shady Shehata. 2024. [Desert camels and oil sheikhs: Arab-centric red teaming of frontier LLMs](#). *Preprint*, arXiv:2410.24049.

- Evan Shieh, Faye-Marie Vassel, Cassidy R. Sugimoto, and Thema Monroe-White. 2026. [Intersectional biases in narratives produced by open-ended prompting of generative language models](#). *Nature Communications*, 17(1):1243.
- Bethany Shockley and Justin J. Gengler. 2024. [Sharing citizenship: Economic competition, cultural threat, and immigration preferences in the rentier state](#). *Political Science Research and Methods*, 12(1):59–75.
- Lex Thijssen, Marcel Coenders, and Bram Lancee. 2021. [Ethnic discrimination in the Dutch labor market: Differences between ethnic minority groups and the role of personal information about job applicants—evidence from a field experiment](#). *Journal of International Migration and Integration*, 22(3):1125–1150.
- Kåre Vernby and Rafaela Dancygier. 2019. [Can immigrants counteract employer discrimination? a factorial field experiment reveals the immutability of ethnic hierarchies](#). *PLOS ONE*, 14(7):e0218044.
- Junting Ye, Shuchu Han, Yifan Hu, Baris Coskun, Meizhu Liu, Hong Qin, and Steven Skiena. 2017. [Nationality classification using name embeddings](#). In *Proceedings of the 2017 ACM Conference on Information and Knowledge Management (CIKM '17)*, pages 1897–1906. ACM.
- Pardis Sadat Zahraei, Gokhan Tur, Dilek Hakkani-Tür, and Ehsaneddin Asgari. 2025. [The alignment veto: How safety training suppresses cultural knowledge in LLMs](#). *Preprint*, arXiv:2510.13154.